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Cretaceous paleomagnetism of the eastern South China Block: establishment of the stable body of SCB

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Abstract

A paleomagnetic investigation was performed on the Cretaceous red sandstones collected at the eastern side of the South China Block (SCB), China, and attempted to establish the stable part of the SCB since the Cretaceous. Paleomagnetic specimens were collected at 39, 25 and 14 sites from three independent parts: the northern, central and southern regions of eastern SCB, respectively. Characteristic directions of higher temperature components (HTCs) with an unblocking temperature of ~ 680 °C were isolated from 69/78 sites. The optimal concentrations of global mean HTC directions calculated using the direction-correction tilt test were achieved at 79 ± 19%, 95 ± 27%, 71 ± 37% and 117 ± 98% untilting for the Early Cretaceous sites from the northern part, Late Cretaceous sites from the northern, central and southern parts, respectively. This treatment gives positive tilt tests or brings the optimal concentration not far from being indistinguishable from positive tilt tests, although this observation can be due to imperfect separation of a primary component (HTC) from a secondary one (lower temperature component). We adopted completely (100%) untilted directions of the HTCs as the paleomagnetic field directions during the Cretaceous, because the mean directions after complete untilting were almost equal to each mean direction after incomplete untilting showing the optimal concentration. The mean paleomagnetic poles for three independent parts were located at almost the same positions and were indistinguishable from that for Sichuan, the western side of SCB at the 95% confidence level. This observation indicates that there is no relative movement between the eastern and western sides of SCB and implies that a large part of the SCB (excluding a 400-km-wide swath along the Red River Fault) has behaved as its coherent (stable) body since the Cretaceous. The Cretaceous paleomagnetic pole for the stable body of the SCB (latitude = 80.0°N, longitude = 206.7°E, A_{95} = 2.5°) is worth using as a reference Cretaceous paleomagnetic pole of the SCB and agrees strongly with the reference Eurasian paleomagnetic pole during the Cretaceous period.
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Keywords: paleomagnetic pole; South China Block (SCB); Cretaceous; red sandstones; tectonics

1. Introduction

Since the Mesozoic, Southeast Asia has undergone tectonic deformation due to the collision of India with Asia [1]. Paleomagnetic analyses have demonstrated tectonic movement of several blocks and regions located in Southeast Asia: southeastward translation (extrusion) and clockwise rotation of the Khorat plateau in the Indochina block [2], clockwise rotation [3–6] and southward displacement [6] of the Simao terrane in the Indochina block, and clockwise

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rotation of a region near the Indus–Zangbo suture [7,8] (Fig. 1, marked by characters *m* to *t*). These deformed blocks and regions are located outside of the South China Block (SCB) and near the Jinsha suture–Red River Fault (RRF).

Paleomagnetic studies of the SCB have revealed that tectonic deformation due to the India–Asia collision is evident in the interior part of the SCB. At first, Otofuji et al. [7] suggested that Sichuan province, on the western side of the SCB, had behaved as a tectonically separate block within the SCB since the Cretaceous. They concluded that the Sichuan province had translated northward by about 500 km with respect to the main part of the China block by comparing it with the Cretaceous paleomagnetic pole (latitude = 70°N, longitude = 208°E, $A_{95} = 5°$) of the China block [19].

After that, Enkin et al. [20] critically reviewed all available paleomagnetic data of Upper Permian to Tertiary age from the main blocks of China. They observed that for almost all periods with sufficient data, the paleomagnetic poles were streaked along a small circle centered on the sampling region. The streaked distribution of paleomagnetic poles indicated that much of China has been affected by small ($< 20°$) differential rotations. They interpreted the rotations as deformation, caused, in part, by the extrusion of the Chinese blocks due to the India–Asia collision. Recently, Otofuji et al. [17] demonstrated the systematically differential rotations in the southwestern part of

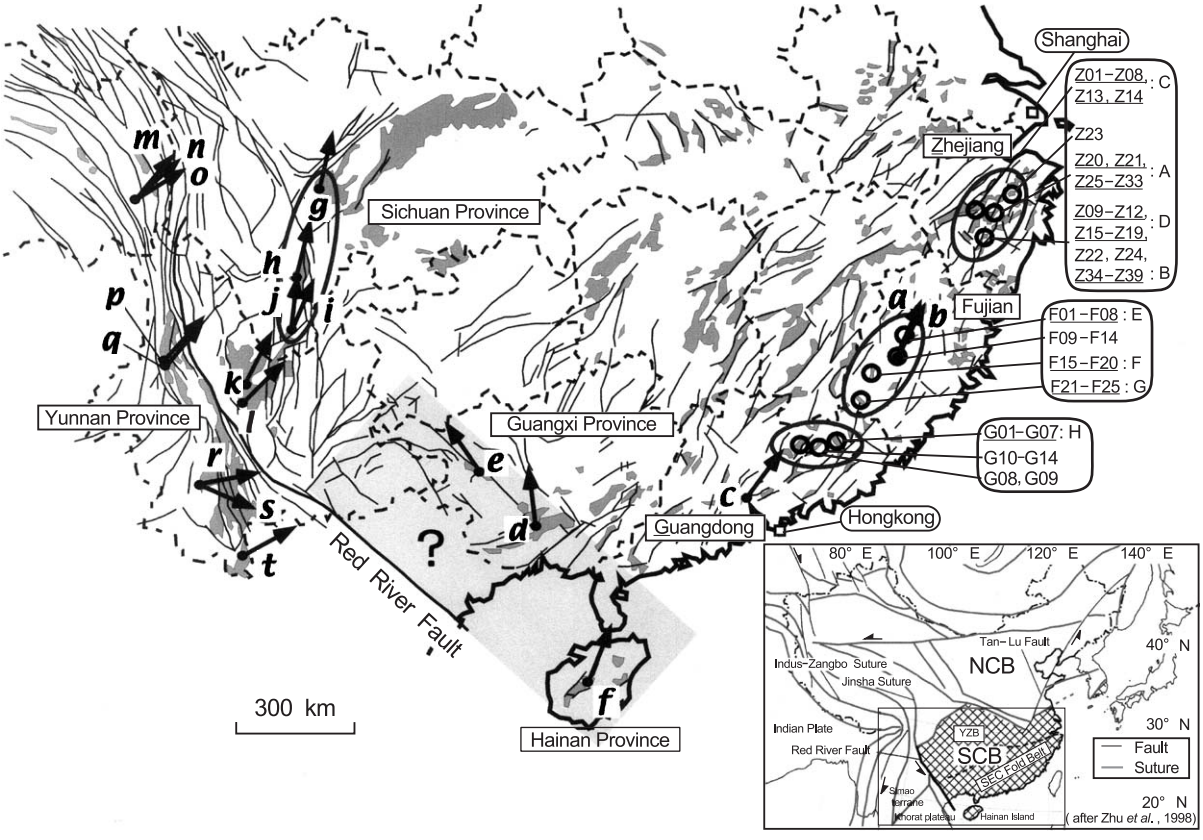


Fig. 1. Small map: tectonic map of East Asia modified from Zhu et al. [9]. Large map: simplified geologic map of South China Block (SCB) and adjacent area showing distribution of faults and Cretaceous basins [10]. Arrows show mean declination reported for the previous studies: *a* [11], *b–e* [12], *f* [13], *g* [14,15], *h* [16], *i*, *j*, *o*, *m* [8], *k* [17], *l* [18], *n* [7], *p* [6], *q* [5], *r*, *t* [3], *s* [4]. Three ellipses of eastern SCB show investigated regions and sampling sites of the present study. Shaded rectangle shows the extruded zone due to India–Asia collision demonstrated by Liu and Morinaga [13].

the SCB near the RRF from the Cretaceous paleomagnetic results [17,18] (Fig. 1, marked by characters *k* and *l*). In Otofuji's study, the amount of rotation was evaluated with respect to 100 Ma pole position of the apparent polar wander path for Eurasia [21].

Liu and Morinaga [13] demonstrated differential southeastward translation (extrusion) of the southern part of the SCB with respect to the suspected stable (coherent) body of the SCB, using Cretaceous paleomagnetic data from Hainan Island (marked by the character *f*) and the Xinlong Formation in Guangxi [12] (marked by the character *d*). They also suggested that a deformed (extruded) range in the interior part of the SCB due to the India–Asia collision is limited to a 400-km-wide swath along the RRF (marked by a shaded rectangle in Fig. 1). They expected the stable body of the SCB since the Cretaceous based on previously reported data from the SCB for Cretaceous paleomagnetic poles. An accurate evaluation of the tectonic deformation within the SCB required paleomagnetic results from the stable body of the SCB. However, reliable paleomagnetic data from the SCB, especially from the eastern side, are still too scarce to construct a reference Cretaceous paleomagnetic pole for the coherent (stable) body of the SCB since the Cretaceous.

The present investigation aims to present the Cretaceous paleomagnetic data from Zhejiang Province and to define a reference Cretaceous paleomagnetic pole of the eastern side of the SCB, reintroducing previously reported data from the Fujian and Guangdong Provinces [22] and the Zhejiang Province [23]. Finally, we propose a more reliable coherent (stable) body of the SCB, which has not suffered tectonic deformation due to the collision of India with Asia since the Cretaceous, and its reference Cretaceous paleomagnetic pole.

2. Geological setting, sampling and laboratory procedures

The South China Block, located in southeast China, is divided into two major fragments, one is the Yangtze craton (YZB) and the other is an exotic terrane named the Southeast China (SEC) Fold Belt or the Huanan block (small map in Fig. 1). Although several different interpretations of the timing of the unification of the two fragments are presented [24–32], the unification is considered to have been complete by the middle Mesozoic at the latest [30]. Therefore, a discussion of the tectonic evolution of the SCB since the Cretaceous does not require a distinction to be made between the two fragments.

Most sedimentary basins on the eastern side of the SCB were controlled by fault movement (extensional or pull-apart basins) (Fig. 1). Within the eastern side of SCB, northeast–southwesterly trending faults are dominant, considered to have been active in the Mesozoic, especially during the Cretaceous period [33]. The rocks of red or purplish-red, fine- to coarse-grained sandstones, and conglomerates in these basins were deposited in a fluviolacustrine environment. Red beds have highly stable natural remanent magnetization, and paleomagnetic data from red beds are very useful for the determination of paleomagnetic poles. Therefore, the Cretaceous red beds distributed around the eastern side of the SCB are the most useful materials for detecting tectonic deformation within this area since the Cretaceous age.

Fig. 1 shows the investigated basins located at the eastern side of the SCB, where we collected paleomagnetic samples for this study. We collected core samples from 20 and 19 sites in the Early and the Late Cretaceous basins, respectively, located at the northern part of the studied range (Zhejiang Province).

In Zhejiang Province, Guantou, Chaochuan, Fangyan and Tangsheng Formations from the Early Cretaceous, are classified in order from the bottom. The Guangtou Formation varies from 40 to 2300 m in thickness and consists of purplish-red or grayish green, fine- to coarse-grained sandstones and siltstones. Fossils identified in the formation include vertebrates (*Paralycoptera wui*, *Yongkangichthys hsitanensis* and *Chetungichthys brevicephalus*), pelycepods (*Trigonioides yongkangensis*, *Plicatounio multiplicatus* and *Nipponia zhejiangensis*) and gastropods (*Brotiopsis wakinoensis*, *Brotiopsis multicostata* and *Galba yongkangensis*), indicative of Early Cretaceous time [34]. We collected core samples at one site (Z24) in this formation.

The Chaochuan Formation varies from 500 to 1350 m in thickness and consists of purplish-red, fine- to medium-grained tuffaceous sandstones and siltstones. Fossils are rarely found in the formation and include pelycepods (*Trigonioides subscalaris*, *Plicatounio* cf.

naktongesis and *Nakamuranaia chingshanensis*) and gastropods (*Lioplacodes* aff. *Cholnokyi*, *Campeloma jousseaume* and *Viviparus onogoensis*), indicative of Early Cretaceous time [34]. The radiometric age of the volcanic rock (tuff) intercalated in this formation (118.6 Ma by Rb–Sr method), also supports Early Cretaceous age [34]. We collected core samples at 12 sites (Z21 to Z23, Z25 to Z33) in this formation.

Core samples were collected from one site (Z20) in the Fangyan Formation, with a thickness of about 200 m. The formation consists of purplish-red or red, fine-grained sandstones. This formation is conformably laid over the Chaochuan Formation, but no fossils are described in the literature [34].

The Tangsheng Formation varies from 70 to 2350 m in thickness and consists of purplish-red or red, fine- to medium-grained sandstones and siltstones. Plant fossils identified rarely in the formation include *Psseudofrenelopsis* (*P.*) *parceramosa*, *P. papillosa* and *Pagiophyllum* sp., indicative of Early Cretaceous time [34]. The radiometric age of the volcanic rock intercalated in this formation (110.9 Ma by K–Ar method), also supports Early Cretaceous age [34]. Core samples were collected at six sites (Z34 to Z39) in this formation.

In Zhejiang Province, Liangtoutang and Quxian Formations, from the Late Cretaceous, are classified in order from the lower part. The Liangtoutang Formation varies from 175 to 690 m in thickness and consists of purplish-red, fine- to coarse-grained sandstones and siltstones, and is conformably laid over the Tangsheng Formation. The Quxian Formation consists of purplish-red or red, fine- to coarse-grained sandstones and siltstones. The thickness of the formation is generally over 1000 m and its maximum is 2360 m. The presence of fossil dinosaurs eggs in both formations indicates Late Cretaceous age [34]. Core samples were collected at 9 and 10 sites in the Liangtoutang and Quxian Formations, respectively.

We collected core samples from 25 and 14 sites in the Late Cretaceous basins located at the central and southern parts of the studied range (Fujian and Guangdong Provinces), respectively. In the Fujian Province, Shaxian and Chongan Formations, from the Late Cretaceous, are classified in order from the bottom. The Shaxian Formation consists of purplish-red, red or grayish white, fine- to coarse-grained sandstones and siltstones and the Chongan Formation consists of pink, medium- to coarse-grained sandstones containing red sandstone pebbles. We collected

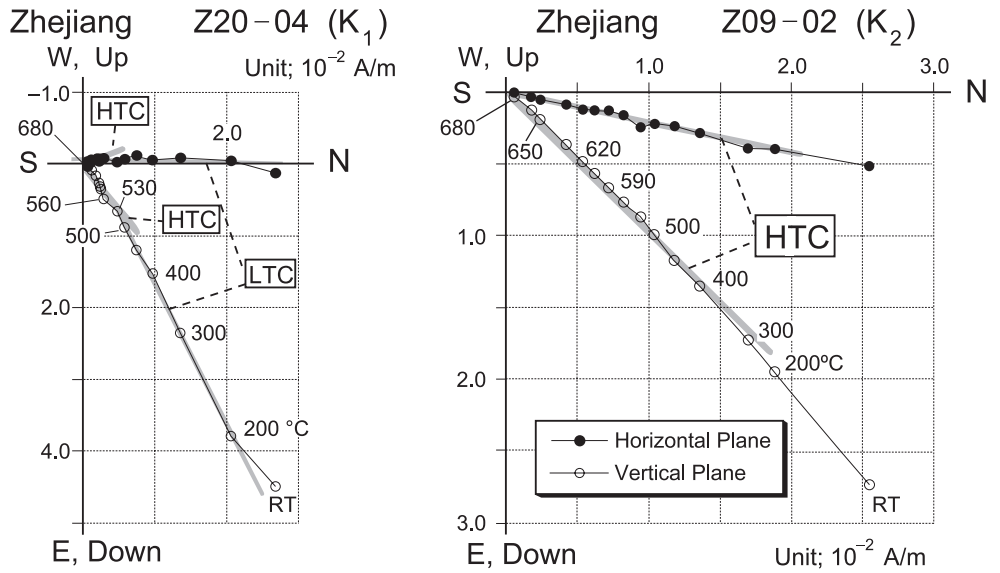


Fig. 2. Representative plots of progressive thermal demagnetization results of red sandstone specimens from the northern part of the studied range (Zhejiang Province); orthogonal vector projections on to the horizontal plane (solid circles) and the N–S vertical plane (open circles). Numbers adjacent to symbols indicate temperature steps (°C). In situ (geographic) coordinates.

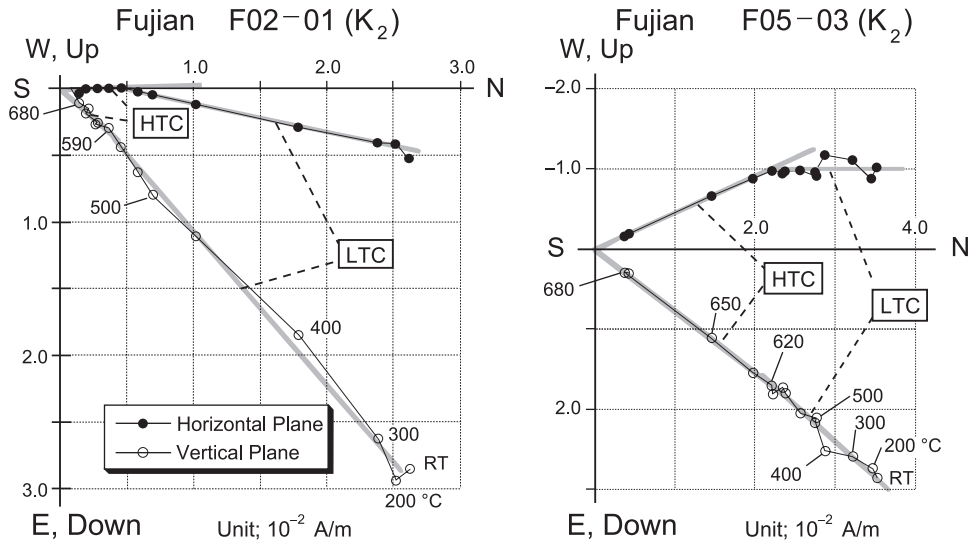


Fig. 3. Representative plots of progressive thermal demagnetization results of red sandstone specimens from the central part of the studied range (Fujian Province); orthogonal vector projections on to the horizontal plane (solid circles) and the N–S vertical plane (open circles). Numbers adjacent to symbols indicate temperature steps (°C). In situ (geographic) coordinates.

core samples at 20 and 5 sites in the Shaxian and Chongan Formations, respectively. For Guangdong Province, only the Dengta Formation is classified as Late Cretaceous. The Dengta Formation consists of pink, fine- or coarse-grained sandstones. We collected core samples at 14 sites in this formation. Detailed information regarding the age and thickness of the Shaxian, Chongan and Dengta Formations could not be obtained, thus the age of these formations is based on the geological map published by the Ministry of Geology and Mineral Resources of China [10].

The latitudes and longitudes of the sites were determined using a portable navigational instrument (SONY PYXIS). Paleomagnetic cores were sampled

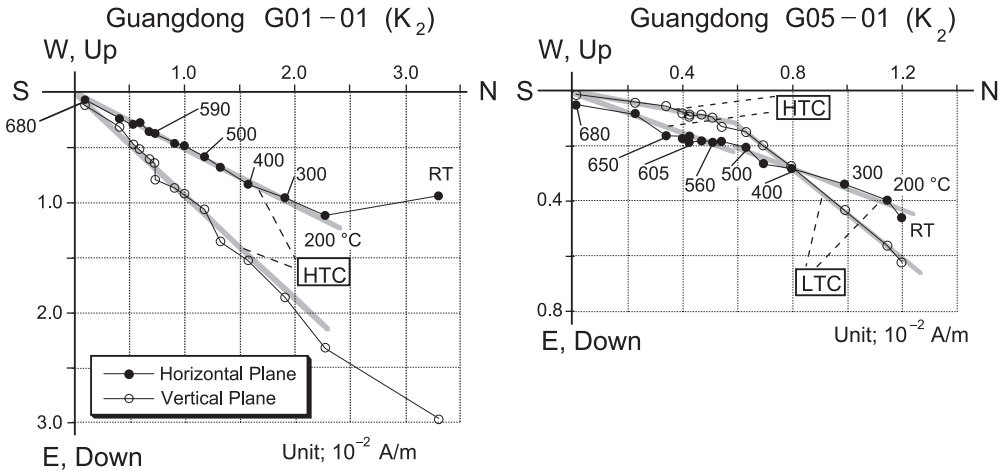


Fig. 4. Representative plots of progressive thermal demagnetization results of red sandstone specimens from the southern part of the studied range (Guangdong Province); orthogonal vector projections on to the horizontal plane (solid circles) and the N–S vertical plane (open circles). Numbers adjacent to symbols indicate temperature steps (°C). In situ (geographic) coordinates.

Table 1
Paleomagnetic results for Early Cretaceous red sandstones from the northern part of the studied range (Zhejiang Province)

The northern part of the studied range (Zhejiang Province)																		
Early Cretaceous (K1)																		
Site No.	Site		Bedding		*N**	In situ		Untilted (79%)		Untilted (100%)		*k*	α95 (°)	VGP#		Lithology	Formation	Reference
	Latitude (°N)	Longitude (°E)	Strike (°)	Dip (°)		Declination (°)	Inclination (°)	Declination (°)	Inclination (°)	Declination (°)	Inclination (°)			Latitude (°N)	Longitude (°E)			
Z20	28.86	120.09	16.0	38.0	6	− 24.2	30.7	− 2.9	46.0	5.4	48.1	62.4	8.5	85.3	205.5	RSS (fine)	Fangyan	
Z21	28.74	120.07	− 156.0	18.0	9	6.8	62.3	− 14.3	55.5	− 18.3	53.1	108.1	5.0	73.6	52.2	RSS (fine)	Chaochuan	
Z22	28.37	119.95	− 168.0	26.0	11	9.7	49.6	− 12.2	44.7	− 16.9	42.3	213.0	3.1	74.4	19.4	RSS (fine)	Chaochuan	
Z23	29.46	120.98	− 22.0	14.0	12	− 2.7	52.0	9.1	47.3	11.7	45.7	249.9	2.8	79.4	220.9	RSS (fine)	Chaochuan	
Z24	28.42	119.86	126.0	30.0	3	no significant number of data										Rss	Guantou	
Z25	28.81	120.09	− 92.0	35.0	7	− 5.9	73.0	− 3.6	45.4	− 3.4	38.0	54.8	8.2	81.9	323.2	Rss	Chaochuan	
Z26	28.97	120.15	− 11.0	9.5	7	12.2	45.9	18.8	42.5	20.3	41.5	140.8	5.1	71.1	221.1	Rss	Chaochuan	
Z27	28.98	120.15	− 10.5	10.0	6	− 8.5	46.9	− 0.2	46.1	1.9	45.6	157.4	5.4	87.4	258.7	Rss	Chaochuan	
Z28	28.98	120.15	− 1.5	10.0	8	− 18.0	48.3	− 9.0	50.0	− 6.6	50.2	226.4	3.7	83.9	51.0	Rss	Chaochuan	
Z29	29.03	120.19	24.0	11.5	7	9.7	48.5	20.1	49.9	23.0	50.0	85.2	6.6	70.0	199.4	Rss	Chaochuan	
Z30	29.06	120.22	− 157.0	13.0	6	4.2	49.3	− 6.0	45.1	− 8.3	43.7	193.7	4.8	81.8	6.6	Rss	Chaochuan	
Z31	29.08	120.22	179.5	17.0	7	11.6	50.5	− 5.1	51.4	− 9.5	51.0	112.5	5.7	81.4	50.3	Rss	Chaochuan	
Z32	29.14	120.23	35.5	12.0	6	− 7.2	45.5	1.2	51.4	3.9	52.8	79.9	7.5	84.6	157.5	Rss	Chaochuan	
Z33	29.21	120.24	158.0	8.0	8	4.6	43.2	− 1.1	45.8	− 2.7	46.3	91.3	5.8	87.1	357.1	Rss	Chaochuan	
Z34	28.43	119.90	− 74.0	35.0	7	48.5	73.6	29.0	47.7	27.5	40.5	64.1	8.4	64.7	215.7	Rss	Tangsheng	23
Z35	28.43	119.90	− 74.0	35.0	7	72.5	72.7	38.9	50.3	36.0	43.5	160.6	4.8	57.9	206.8	Rss	Tangsheng	23
Z36	28.43	119.90	− 74.0	35.0	7	51.2	69.7	32.2	44.4	30.6	37.3	217.6	4.1	61.3	218.3	Rss	Tangsheng	23
Z37	28.43	119.90	− 74.0	35.0	6	56.6	77.7	29.1	52.2	27.3	45.0	121.9	6.1	65.8	207.8	Rss	Tangsheng	23
Z38	28.43	119.90	− 79.0	34.0	9	91.9	73.1	42.2	56.4	37.6	50.1	144.3	5.0	57.4	196.2	Rss	Tangsheng	23
Z39	28.43	119.92	− 95.0	35.5	18	92.3	77.2	21.9	60.9	17.0	54.0	123.7	5.4	74.3	182.8	Rss	Tangsheng	23
Mean	in situ				19 sites	8.5	60.4					15.8	8.7					
	untilted (79%)				19 sites			9.4	50.4			41.8	5.3	81.3	190.1	maximum *k*		
													(k = 25.1, A_{95} = 6.3)					
	untilted (100%)				19 sites					9.6	47.6	37.6	5.5	81.8	207.1			
													(k = 24.9, A_{95} = 6.9°)					

*N**, number of samples having significant components.

VGP#, virtual geomagnetic pole calculated using 100% untilted data.

RSS, red sandstone; Rss, red siltstone.

Table 2
Paleomagnetic results for Late Cretaceous red sandstones from the northern part of the studied range (Zhejiang Province)

The northern part of the studied range (Zhejiang Province)																		
Late Cretaceous (K2)																		
Site No.	Site		Bedding		N*	In situ		Untilted (95%)		Untilted (100%)		k	α95 (°)	VGP#		Lithology	Formation	Reference
	Latitude (°N)	Lon (°E)	Strike (°)	Dip (°)		Declination (°)	Inclination (°)	Declination (°)	Inclination (°)	Declination (°)	Inclination (°)			Latitude (°N)	Longitude (°E)			
Z01	29.07	119.68	− 157.0	13.0	10	9.6	51.5	− 4.0	47.2	− 5.2	47.8	58.3	6.4	85.4	28.4	RSS (fine)	Quxian	
Z02	29.07	119.68	− 135.0	12.0	10	9.6	45.0	1.8	37.7	1.0	38.0	11.5	14.9	82.2	292.8	RSS (fine)	Quxian	
Z03	29.07	119.68	− 150.0	10.0	10	51.4	38.3	43.8	41.2	43.6	42.0	134.7	4.2	51.0	206.2	RSS (fine)	Quxian	
Z04	29.06	119.69	− 178.0	19.0	10	9.4	47.7	− 10.4	46.9	− 11.9	48.0	460.7	2.3	79.6	32.5	RSS (fine)	Quxian	
Z05	29.06	119.69	− 150.0	17.0	10	39.6	37.3	27.0	38.2	26.1	39.4	152.8	3.9	65.6	219.6	RSS (fine)	Quxian	
Z06	29.07	119.72	− 65.0	16.0	10	− 21.9	58.1	− 9.0	46.4	− 9.7	45.2	55.1	6.6	81.1	16.7	RSS (fine)	Quxian	
Z07	29.07	119.72	− 100.0	12.0	10	− 12.3	46.2	− 11.9	34.8	− 12.5	34.2	47.4	7.1	74.6	350.4	RSS (fine)	Quxian	
Z08	29.07	119.72	− 60.0	4.0	10	4.0	57.7	6.4	54.3	6.1	54.0	134.2	4.2	82.5	161.6	RSS (fine)	Quxian	
Z09	28.50	119.97	158.0	16.0	8	16.5	47.5	− 0.1	55.3	− 1.2	55.6	86.3	6.0	82.3	112.7	RSS (fine)	Liangtoutang	
Z10	28.48	119.95	180.0	20.0	4	41.6	50.3	17.5	59.9	15.8	60.2	27.1	18.0	71.9	161.4	RSS (fine)	Liangtoutang	
Z11	28.46	119.91	− 56.0	14.0	11	− 9.3	60.0	2.2	49.4	2.6	48.8	137.1	3.9	87.4	180.0	RSS (fine)	Liangtoutang	
Z12	28.44	119.95	− 116.0	36.0	10	51.3	68.5	5.2	46.4	4.3	44.9	42.4	7.5	85.7	236.1	RSS (fine)	Liangtoutang	
Z13	29.11	119.64	− 38.0	10.0	9	6.8	35.6	10.9	28.7	11.1	28.3	28.6	9.8	72.6	261.2	RMS	Quxian	
Z14	29.11	119.63	− 16.0	6.0	8	8.0	33.6	11.3	31.1	11.4	31.0	24.9	11.3	73.8	257.0	RMS	Quxian	
Z15	28.44	119.90	− 78.5	31.5	9	52.4	76.2	25.2	48.7	24.8	47.2	88.7	9.8	68.2	204.1	Rss	Liangtoutang	23
Z16	28.44	119.90	− 78.5	31.5	7	45.8	71.6	25.8	43.8	25.4	42.3	70.5	8.0	67.0	213.9	Rss	Liangtoutang	23
Z17	28.44	119.90	− 78.5	31.5	9	22.6	81.6	14.1	51.8	14.0	50.2	198.3	3.7	77.6	194.7	Rss	Liangtoutang	23
Z18	28.44	119.90	− 77.0	32.0	14	58.2	77.6	26.7	50.0	26.3	48.4	32.5	8.1	67.0	201.1	Rss	Liangtoutang	23
Z19	28.48	119.95	− 175.0	17.5	8	35.8	38.1	22.2	45.0	21.4	45.2	16.0	17.3	71.0	210.2	Rss	Liangtoutang	23
Mean	in situ				19 sites	19.1	55.8					16.5	8.5					
	untilted (95%)				19 sites			10.7	46.1			36.3	5.6	80.5	212.3	maximum k		
												(k = 29.1, A_{95} = 6.3°)						
	untilted (100%)				19 sites					10.4	45.5	36.2	5.7	81.0	214.3			
												(k = 28.8, A_{95} = 6.4°)						

N, number of samples having significant components.
VGP#, virtual geomagnetic pole calculated using 100% untilted data.
RSS, red sandstone; Rss, red siltstone.

Table 3
Paleomagnetic results for Late Cretaceous red sandstones from the central part of the studied range (Fujian Province)

The central part of the studied range (Fujian Province)																		
Late Cretaceous (K2)																		
Site No.	Site		Bedding		*N**	In situ		Untilted (71%)		Untilted (100%)		*k*	α95 (°)	VGP#		Lithology	Formation	Reference
	Latitude (°N)	Longitude (°E)	Strike (°)	Dip (°)		Declination (°)	Inclination (°)	Declination (°)	Inclination (°)	Declination (°)	Inclination (°)			Latitude (°N)	Longitude (°E)			
F01	26.41	117.79	14.7	10.0	7	15.9	48.6	23.8	48.0	27.0	47.4	46.4	9.0	66.0	196.5	GWSS (medium)	Shaxian	22
F02	26.41	117.79	12.7	22.0	7	14.2	49.5	31.5	46.7	37.5	44.3	825.0	2.1	56.5	199.9	RSS	Shaxian	22
F03	26.41	117.79	14.7	22.0	7	14.3	50.5	32.4	48.1	38.8	45.8	772.5	2.2	55.5	197.5	RSS	Shaxian	22
F04	26.40	117.80	38.7	14.0	6	− 0.9	35.1	5.4	41.0	8.5	43.2	39.9	10.7	82.2	215.2	DGSS (fine)	Shaxian	22
F05	26.40	117.80	48.7	14.0	5	− 13.7	27.6	− 10.7	36.3	− 9.0	39.8	37.2	12.7	81.0	4.9	RSS (fine)	Shaxian	22
F06	26.40	117.79	46.7	12.0	7	− 17.9	35.8	− 14.7	43.4	− 13.0	46.4	79.7	6.8	78.4	37.1	RSS (fine)	Shaxian	22
F07	26.40	117.79	48.7	16.0	7	0.2	33.3	6.3	41.4	9.6	44.5	65.5	7.5	81.4	207.2	RSS (fine)	Shaxian	22
F08	26.40	117.79	40.7	10.0	7	12.5	47.1	19.8	50.1	23.2	51.0	625.5	2.4	69.1	187.7	RSS (fine)	Shaxian	22
F09	25.99	117.38	− 21.3	24.0		only a low temperature component										RSS (medium)	Chongan	22
F10	26.00	117.39	10.7	28.0		only a low temperature component										RSS (coarse)	Chongan	22
F11	26.00	117.39	36.7	26.0	7	12.0	23.2	20.7	29.7	25.0	31.5	30.2	11.2	65.1	223.5	RSS	Chongan	22
F12	26.00	117.39	28.7	20.0		only a low temperature component										RSS (medium)	Chongan	22
F13	26.03	117.42	− 45.3	14.0	4	− 19.7	56.0	− 8.2	50.8	− 4.5	48.2	21.6	20.2	84.9	67.1	RSS (fine)	Chongan	22
F14	26.06	117.44	156.7	28.0	7	36.3	6.5	33.5	23.4	31.1	30.2	56.1	8.1	59.5	220.0	RSS (fine)	Shaxian	22
F15	25.74	116.75	− 157.1	20.0	6	8.2	31.0	0.8	26.5	− 1.8	24.2	230.0	4.4	76.8	304.5	RSS (fine)	Shaxian	22

F16	25.73	116.75	− 171.1	12.0	5	24.1	40.4	16.8	42.1	13.6	42.5	73.3	9.0	77.6	209.0	RSS (fine)	Shaxian	22
F17	25.73	116.75	128.9	14.0	4	29.0	36.5	27.4	46.3	26.4	50.2	25.4	18.6	66.2	187.9	RSS (fine)	Shaxian	22
F18	25.73	116.75	− 27.1	10.0	7	4.0	47.6	10.0	43.6	12.1	41.8	312.3	3.4	78.9	212.7	RSS (fine)	Shaxian	22
F19	25.72	116.75	154.9	18.0	7	18.8	26.2	13.1	34.6	10.1	37.7	50.4	8.6	79.7	231.0	RSS (fine)	Shaxian	22
F20	25.72	116.75	131.9	16.0	5	8.4	42.3	1.1	51.4	− 3.2	54.8	130.3	6.7	80.0	101.5	RSS (coarse)	Shaxian	22
F21	25.06	116.44	115.0	20.0	8	19.1	32.8	17.7	46.9	16.8	52.7	112.6	5.2	73.2	173.2	RSS (fine)	Shaxian	22
F22	25.05	116.44	− 147.0	20.0	6	15.5	37.0	6.5	31.6	3.5	28.9	51.5	9.4	79.8	277.0	RSS (fine)	Shaxian	22
F23	25.05	116.44	171.0	20.0	6	24.3	39.3	12.6	45.9	6.8	47.8	296.2	3.9	82.8	172.7	RSS (fine)	Shaxian	22
F24	25.05	116.44	− 171.0	20.0	4	21.7	53.4	2.2	54.1	− 5.6	53.1	60.7	11.9	80.1	88.2	RSS (fine)	Shaxian	22
F25	25.05	116.43	− 173.0	28.0	7	26.4	42.4	7.1	45.9	− 1.2	45.3	470.8	2.8	87.9	85.1	RSS (fine)	Shaxian	22
Mean	in situ				22 sites	11.9	39.2					25.3	6.3					
	untilted (71%)				22 sites			11.7	43.0			38.4	5.1	79.5	206.9	maximum k		
													($k = 34.7$, $A_{95} = 5.3°$)					
	untilted (100%)				22 sites					11.5	44.2	35.3	5.3	79.7	201.7			
													($k = 29.7$, $A_{95} = 5.8°$)					

N, number of samples having significant components.

VGP#, virtual geomagnetic pole calculated using 100% untilted data.

RSS, red sandstone.

GWSS, grayish-white sandstone; DGSS, dark-gray sandstone.

using a portable gasoline-powered drill, and were oriented with a magnetic compass. Remanent magnetization of specimens was measured on a Natsuhara Giken SMM-85 spinner magnetometer. Progressive thermal demagnetization was adopted for eliminating viscous and unstable magnetic components and isolating higher and lower temperature components. The direction of characteristic remanent magnetization (ChRM) direction was computed using principal component analysis [35].

3. Results of magnetic measurements and interpretation

3.1. Progressive thermal demagnetization

Collected sandstone and siltstone specimens generally show stable behavior through the thermal treatment (Figs. 2–4). However, medium- or coarse-grained sandstones from the southern parts of the studied range (sites G10 to G14) show very unstable behavior and have no significant magnetic component. Most specimens show two stable magnetization phases, with two different unblocking temperatures, the lower temperature component (LTC) at ~ 590 °C, and the higher temperature component (HTC) at ~ 680 °C. Most LTCs do not trend toward the origin, while the HTCs show the trend (Fig. 2, left; Figs. 3 and 4, right). The high unblocking temperature of the remanence indicates that it is carried by hematite. For specimens from several sites, the directions of the two different unblocking temperature phases are almost equal (Fig. 2, right; and Fig. 4, left). Therefore, they cannot be separated using the principal component analysis. Specimens from three sites in the central part of the studied range (Fujian sites, F09 F10 and F12) show only a LTC.

3.2. Determination of paleomagnetic directions

We regard the HTC direction as the ChRM direction of the specimens, which is useful for paleomagnetic analysis. On the other hand, the LTC of the specimens can be regarded as a secondary component, acquired after tilting of the sediment layer occurs. The mean of in situ LTC directions shows a higher concentration than that of completely untilted directions. For the instance, the concentration ($k = 60.6$) of the in situ LTC direction mean for 14 sites (Z20 to Z33) of Early Cretaceous age from the northern part, whose specimens show two stable magnetization phases of LTC and HTC with slightly different directions to each other, is three times higher than that of the 100% untilted direction mean (20.7).

The studied range was divided into three independent parts (ellipses in Fig. 1): the northern (Zhejiang Province), central (Fujian Province) and southern (Guangdong Province) parts. This geographical grouping of paleomagnetic data is adopted in order to prepare enough data sets to average out paleomagnetic secular variation and to discuss the boundary and extent of the stable body of the SCB.

All the paleomagnetic results are listed in Tables 1–4 and the site mean directions are shown in Figs. 5–8, Early (K1) and Late Cretaceous (K2) in the northern part (Zhejiang), K2 in the central part (Fujian) and southern part (Guangdong), respectively. In situ (geographical) mean directions of several sites, especially for K1 sites in the northern part (Fig. 5, left), are fairly parallel to the geocentric axial dipole field. However, the strata at these sites are more gently tilted (less than 15°) than at the other sites; thus, we believe the HTC's are not secondary magnetizations acquired after tilting.

The optimal concentration calculated using the direction-correction (DC) tilt test [36] is achieved at $79 \pm 19\%$, $95 \pm 25\%$, $71 \pm 37\%$ and $117 \pm 98\%$ untilting for K1 in the northern part (except for Z24, because of a small number of original data (only three data points)), K2 in the northern, central and southern parts, respectively. Although the bedding attitudes are not steep enough for a strong tilt test, especially for K2 in the southern part, this treatment gives positive tilt tests or brings the optimal concentrations not far from being indistinguishable from the positive tilt test. Thus, we interpret the HTCs to be near-primary origin before tilting. The optimal concentration at an incomplete untilting may be regarded as an apparent phenomenon brought on by imperfect separation of a primary component (HTC) from a secondary one (LTC).

As the mean directions obtained through complete (100%) untilting are almost equal to each mean direction after incomplete untilting showing the optimal concentration, we adopt the completely untilted

Table 4
Paleomagnetic results for Late Cretaceous red sandstones from the southern part of the studied range (Guangdong Province)

The southern part of the studied range (Guangdong Province)																			
Late Cretaceous (K2)																			
Site No.	Site		Bedding		*N**	In situ		Untilted (100%)		Untilted (117%)		*k*	α95 (°)	VGP#		Lithology	Formation	Reference
	Latitude (°N)	Longitude (°E)	Strike (°)	Dip (°)		Declination (°)	Inclination (°)	Declination (°)	Inclination (°)	Declination (°)	Inclination (°)			Latitude (°N)	Longitude (°E)			
G01	24.24	115.74	165.3	16.0	7	24.7	38.5	12.1	47.3	9.4	48.5	136.8	5.2	78.4	181.9	DRSS (fine)	Dengta	22
G02	24.24	115.74	165.3	16.0	7	20.7	32.4	10.4	40.5	8.2	41.6	84.5	6.6	80.4	210.3	DRSS (fine)	Dengta	22
G03	24.14	115.81	147.3	26.0	7	14.3	30.7	− 2.5	47.3	− 6.9	49.3	214.0	4.1	85.1	88.9	RSS (fine)	Dengta	22
G04	24.14	115.81	151.3	26.0	7	15.5	17.4	5.6	34.1	3.0	36.5	96.5	6.2	82.5	251.0	RSS (fine)	Dengta	22
G05	24.14	115.81	139.3	18.0	6	9.7	16.6	4.4	30.0	3.1	32.1	31.1	12.2	81.0	267.8	RSS (fine)	Dengta	22
G06	24.14	115.80	131.3	14.0	7	14.3	28.5	9.5	40.7	8.4	42.7	122.6	5.5	81.3	209.6	RSS (fine)	Dengta	22
G07	24.14	115.80	105.3	12.0	7	20.0	22.2	20.6	34.2	20.7	36.2	170.2	4.6	70.1	217.5	RSS (fine)	Dengta	22
G08	24.05	114.80	115.4	5.0	6	20.3	52.3	19.6	57.3	19.5	58.1	81.7	7.5	68.3	160.5	RSS (fine)	Dengta	22
G09	24.10	114.77	129.4	8.0	6	26.1	55.2	22.6	62.9	21.8	64.2	331.0	3.7	62.6	151.5	RSS (fine)	Dengta	22
G10	24.09	115.20	− 128.6	32.0		no significant component										RSS (coarse)	Dengta	22
G11	24.09	115.20	− 128.6	32.0		no significant component										RSS (coarse)	Dengta	22
G12	24.09	115.20	− 118.6	30.0		no significant component										RSS (coarse)	Dengta	22
G13	24.09	115.20	− 92.6	22.0		no significant component										RSS (medium)	Dengta	22
G14	24.09	115.20	− 110.6	26.0		no significant component										RSS (coarse)	Dengta	22
Mean	in situ				9 sites	17.8	32.7					31.1	9.4					
	untilted (100%)				9 sites			10.7	44.0			42.8	8.0	79.8	188.3			
												(k=46.1, A_{95}=7.7°)						
	untilted (117%)				9 sites					8.9	45.8	43.0	7.9	80.8	177.7	maximum *k*		
												(k=41.9, A_{95}=8.0°)						

N, number of samples having significant components.
VGP#, virtual geomagnetic pole calculated using 100% untilted data.
RSS, red sandstone; DRSS, dark red sandstone.

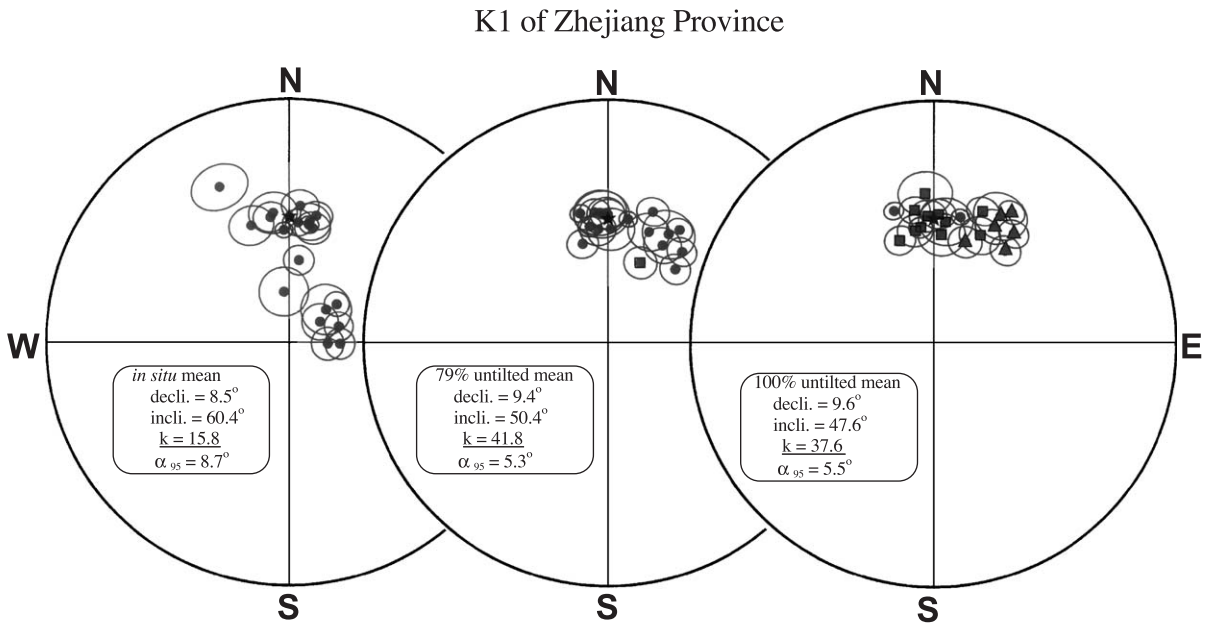


Fig. 5. Equal-area projection of the site mean directions for Early Cretaceous red sandstones from the northern part of the studied range (Zhejiang Province); in situ (left), 79% (center) and 100% (right) untilted directions. (★) The geocentric axial dipole field direction. (■) The mean directions of sites in an identical basin 'A'. (▲) The mean directions of sites in an identical basin 'B'.

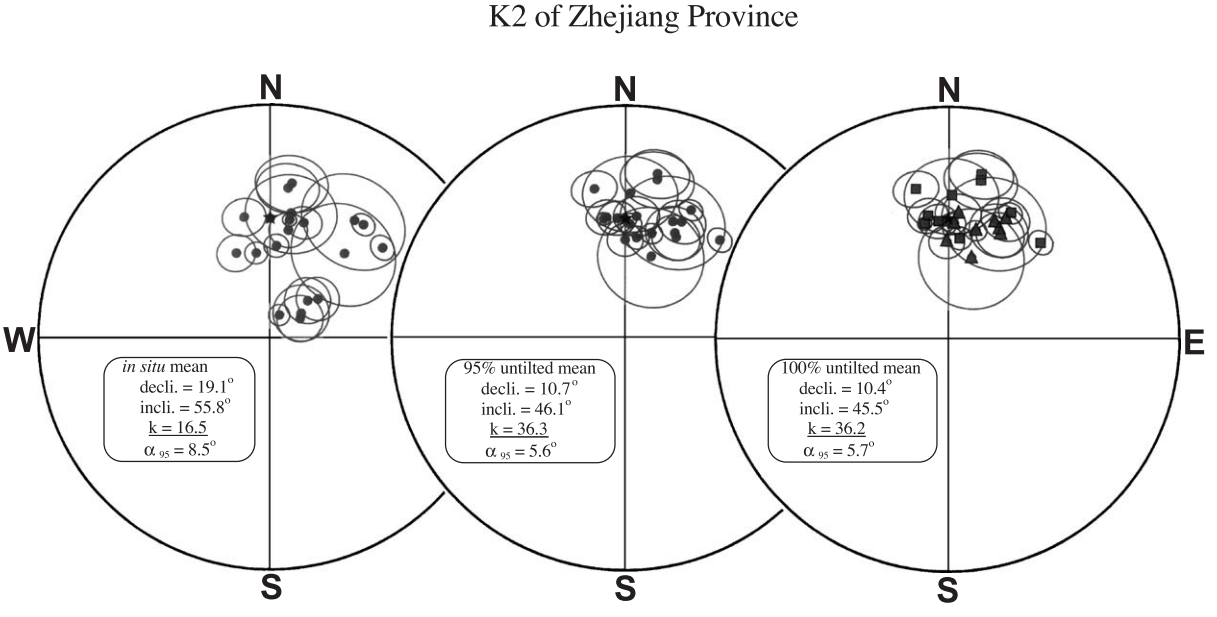


Fig. 6. Equal-area projection of the site mean directions for Late Cretaceous red sandstones from the northern part of the studied range (Zhejiang Province); in situ (left), 95% (center) and 100% (right) untilted directions. (★) The geocentric axial dipole field direction. (■) The mean directions of sites in an identical basin 'C'. (▲) The mean directions of sites in an identical basin 'D'.

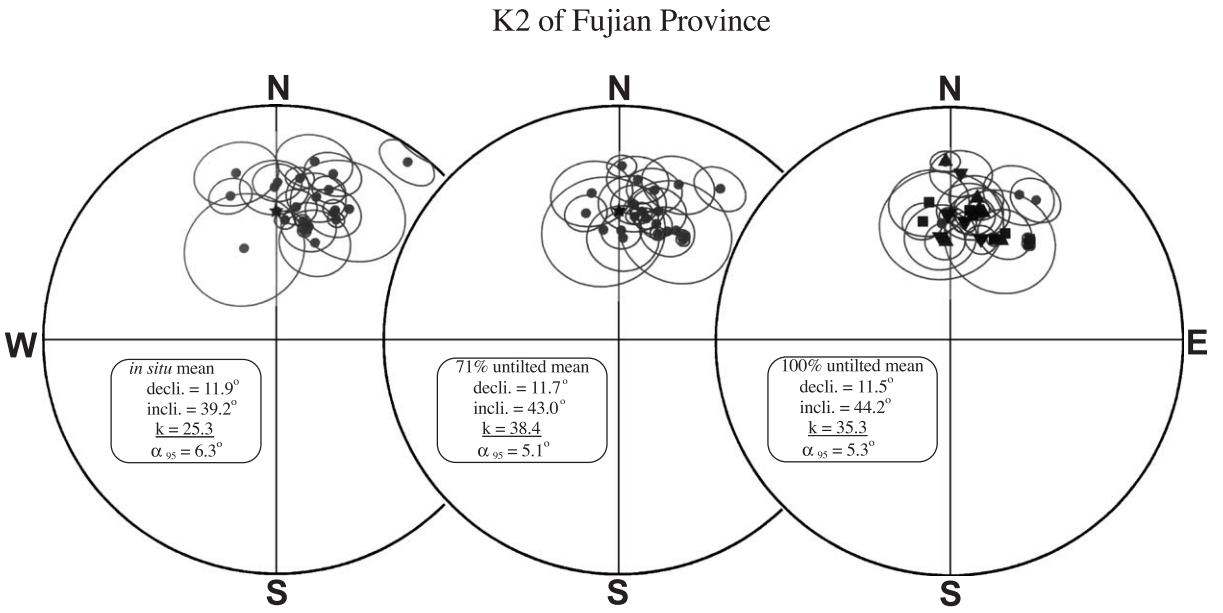


Fig. 7. Equal-area projection of the site mean directions for Early Cretaceous red sandstones from the central part of the studied range (Fujian Province); in situ (left), 71% (center) and 100% (right) untilted directions. (★) The geocentric axial dipole field direction. (■) The mean directions of sites in an identical basin 'E'. (▲) The mean directions of sites in an identical basin 'F'. (▼) The mean directions of sites in an identical basin 'G'.

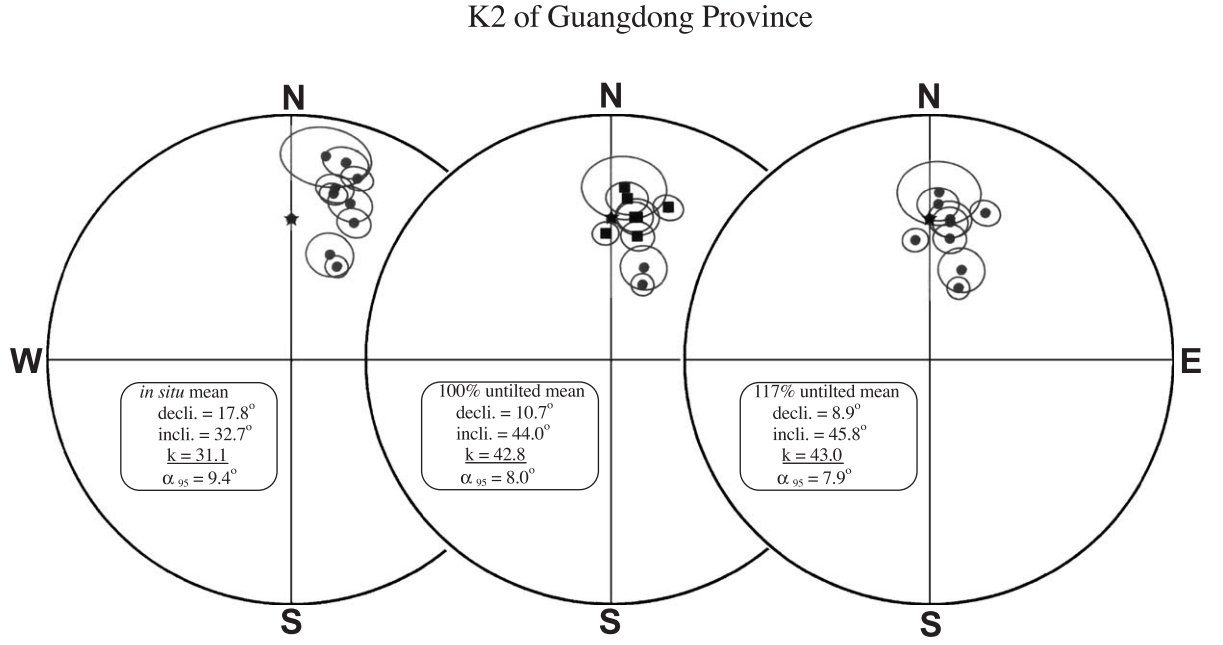


Fig. 8. Equal-area projection of the site mean directions for Early Cretaceous red sandstones from the southern part of the studied range (Guangdong Province); in situ (left), 100% (center) and 117% (right) untilted directions. (★) The geocentric axial dipole field direction. (■) The mean directions of sites in an identical basin 'H'.

directions as the paleomagnetic field directions during the Cretaceous.

4. Discussion

Paleomagnetic poles for each sampling site were calculated using the mean directions of the completely untilted data. Fig. 9 shows the mean paleomagnetic poles of each basin with over five sites (Fig. 1); A (Z20, Z21, Z25 to Z33), B (Z34 to Z39), C (Z01 to Z08, Z13, Z14), D (Z09 to Z12, Z15 to Z19), E (F01 to F08), F (F15 to F20), G (F21 to F25) and H (G01 to G07). The site means of the completely untilted directions are indicated using different symbols for the independent basins A to H, in Figs. 5–8. Streaked distribution of mean paleomagnetic poles, indicative of vertical axis rotations, demonstrated by a previous compilation of paleomagnetic poles [20] is not obvious, with the exception of one paleomagnetic pole (from the Lishui basin, B in Fig. 9). Although one of the authors (Liu) stated regional tectonic deformation in the Lishui basin [23], we now think that the deflection of this paleomagnetic pole is brought on by insufficient time averaging because sampling was carried out in the small area.

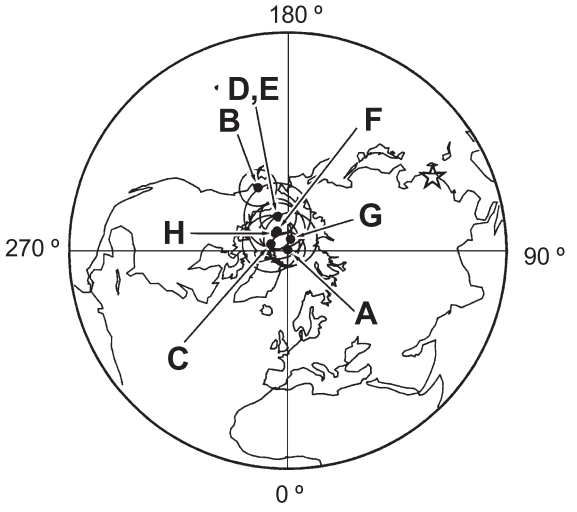


Fig. 9. Mean paleomagnetic poles of each basin with over five sites obtained from the present study; A (Z20, Z21, Z25 to Z33), B (Z34 to Z39), C (Z01 to Z08, Z13, Z14), D (Z09 to Z12, Z15 to Z19), E (F01 to F08), F (F15 to F20), G (F21 to F25) and H (G01 to G07). A star indicates the studied area.

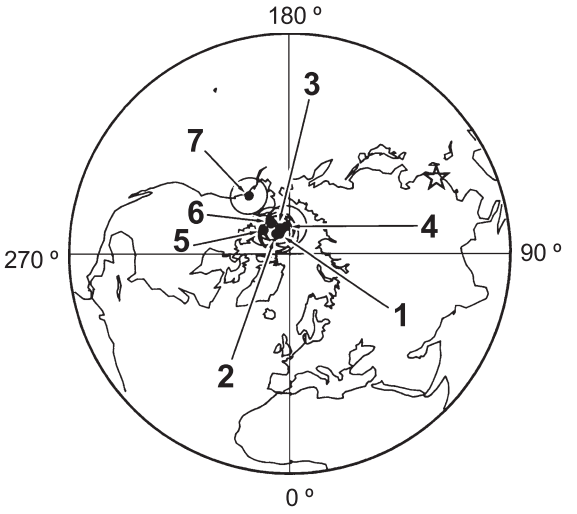


Fig. 10. Cretaceous Paleomagnetic poles from South China: (1) K1 of the northern part of the studied range (this study); (2) K2 of the northern part of the studied range (this study); (3) K2 of the central part of the studied range (this study); (4) K2 of the southern part of the studied range (this study); (5) K1 of Zhejiang (Lin [37]); (6) K of Sichuan, the western side of SCB (compiled by Gilder et al. [12]); (7) K of coastal provinces of SCB (compiled by Gilder et al. [12]). A star indicates the studied area.

The global mean paleomagnetic poles calculated for the three independent parts are as follows: for K1 in the northern part, latitude = 81.8°N, longitude = 207.1°E, $k = 24.9$, $A_{95} = 6.9°$; for K2 in the northern part, latitude = 81.0°N, longitude = 214.3°E, $k = 28.8$, $A_{95} = 6.4°$; for K2 in the central part, latitude = 79.7°N, longitude = 201.7°E, $k = 29.7$, $A_{95} = 5.8°$; for K2 in the southern part, latitude = 79.8°N, longitude = 188.3°E, $k = 46.1$, $A_{95} = 7.7°$ (Fig. 10). These four Cretaceous paleomagnetic poles from the present study are located at almost the same positions.

Lin [37] reports a mean paleomagnetic pole (latitude = 77.1°N, longitude = 227.6°E, $A_{95} = 5.5°$) for the Early Cretaceous red beds collected from the seven sites, situated in the northern area of Zhejiang. This pole and the four VGP positions are indistinguishable from one another at a 95% confidence level (Fig. 10). Therefore, no relative tectonic movement since the Cretaceous is paleomagnetically detected among the studied regions of the SCB.

In contrast, Gilder et al. [12] combine their results from Yongan (Shaxian Formation) in Fujian and the

Sanshui Formation in Guangdong with previously reported results from the Hekou Formation in Fujian [11] and from Guangdong [38], and calculate a mean Late Cretaceous paleomagnetic pole (latitude = 63.7°N, longitude = 215.0°E, A_{95} = 6.7°) in the coastal provinces of Fujian and Guangdong. The pole position for the coastal provinces differs significantly from both paleomagnetic poles for the central (Fujian) and southern (Guangdong) parts defined in the present study (Fig. 10). They describe a possibility corresponding to a coherent block rotation (12 ± 8° clockwise) of the coastal region with respect to Sichuan, the western side of the SCB. However, they also state that the number of sampling sites is small and that most of the previous studies are preliminary. The Shaxian and Hekou Formations are the same formation as we collected at the F09 to F14 sampling sites. Although a mean paleomagnetic pole was obtained by Zhai et al. [11] using paleomagnetic results from seven sites, the remaining three mean paleomagnetic poles [12,38] are calculated using insufficient data from one or two sites. Thus, we consider the mean paleomagnetic pole for the coastal region to be less reliable because paleomagnetic secular variation is not fully averaged out.

Gilder et al. [12] also compiled previously reported paleomagnetic results from the Sichuan Province, on the western side of the SCB [8,14–16] and calculate a mean Cretaceous paleomagnetic pole (latitude = 76.9°N, longitude = 211.6°E, A_{95} = 6.0°). The paleomagnetic pole for Sichuan is indistinguishable from those for the eastern side of the SCB at a 95% confidence level (Fig. 10). This observation indicates that there is no relative movement between the western and eastern parts of SCB. In addition, this implies that a large part of the SCB, excluding some regions nearer to the RRF, has behaved as a coherent body since the Cretaceous.

In some regions nearer to the RRF within the SCB, differential rotation of tectonic blocks situated at the southern part of Sichuan and the eastern part of Yunnan [17] and differential southeastward translation (extrusion) of Hainan Island and Guangxi region [13] are demonstrated by the Cretaceous paleomagnetic investigations. Otofuji et al. [17] identified a systematically differential rotation, a long distance from the RRF, which is a part of the deformation mode of the Asian continent due to the collision with Asia. Liu and Morinaga [13] state that a limited area, including Hainan Island and southwest Guangxi, less than 400 km distance northeast the RRF (marked by a shaded rectangle in Fig. 1), where several NW–SE trending faults parallel to the RRF are dominant, was dissected by a fault and differentially translated southeastward by the stress due to the indentation of India into Asia. The results of the present study also support the idea that the effects of the India–Asia collision did not extend over the northeastern part of the SCB to more than 400 km distant from the RRF [13].

5. Further implications

Cretaceous paleomagnetic studies of the SCB demonstrate that the large part of the SCB (excluding a 400-km-wide swath along the RRF) has behaved as a coherent (stable) body since the Cretaceous. The global mean (latitude = 80.0°N, longitude = 206.7°E, k = 973.5, A_{95} = 2.5°) of five Cretaceous paleomagnetic poles from Sichuan on the western side of the SCB (compiled by Gilder et al. [12]), the northern (two paleomagnetic poles), the central and southern parts of the eastern SCB developed in the present study is worth using as a reference Cretaceous paleomagnetic pole for the stable part of the SCB, when discussing the relative tectonic movements within the SCB, especially in the vicinity of the RRF.

If we apply this reference Cretaceous paleomagnetic pole to treat the tectonic deformation in the vicinity of the RRF quantitatively, the effect of the India–Asia collision can be estimated more precisely. For instance, we can obtain a southward translation of about 9° (8.7 ± 4.0°) and no rotation (0.4 ± 4.5° clockwise) of Hainan Island by comparing the Early Cretaceous paleomagnetic pole (latitude = 77.7°N, longitude = 162.1°E, A_{95} = 4.4°) for Hainan Island [13]. Liu and Morinaga [13] obtained a southward translation of 14° by comparing the Early Cretaceous paleomagnetic pole with that of the suspected coherent part of SCB, and stated that the amount of translation may be overestimated. By establishing the reference Cretaceous paleomagnetic pole of the SCB, we can obtain a more reasonable estimate of the rotation and translation than with the previous estimation. For the Shimao terrane and the Khorat plateau south of the RRF, we can also determine a more

reliable estimate of the rotation and translation against the stable body of the SCB since the Cretaceous through the comparison with this new reference Cretaceous paleomagnetic pole.

Establishment of the reference Cretaceous paleomagnetic pole for the SCB also enables the discussion of the relative movements between SCB and other tectonic units. For the North China Block located north of the SCB, however, the amount of the Cretaceous paleomagnetic data available is small [20]. Therefore, we cannot estimate the relative movement between both blocks at present. We can also discuss the tectonic movements of the SCB with respect to stable Eurasia, excepting China, by comparing the reference Cretaceous paleomagnetic pole of the SCB with a reference Eurasian paleomagnetic pole during the Cretaceous period.

Enkin et al. [20] excluded the Chinese paleomagnetic poles from worldwide selection data [21] and defined Early (latitude = 77.8 °N, longitude = 208.4 °E, A_{95} = 7.0 °) and Late Cretaceous paleomagnetic poles (latitude = 77.9 °N, longitude = 203.8 °E, A_{95} = 14.2 °) for stable Eurasia. These paleomagnetic poles are located at the same positions as the reference Cretaceous paleomagnetic pole for the SCB (Fig. 11). This observation indicates that no relative tectonic movement has been detected between the stable body of the SCB and stable Eurasia since the Cretaceous.

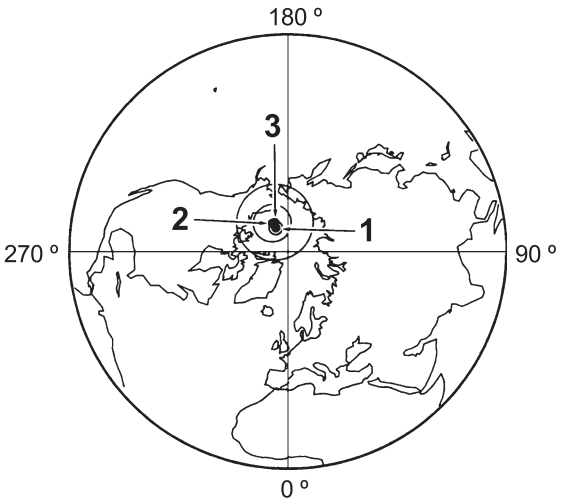


Fig. 11. Cretaceous paleomagnetic poles for SCB from the present study: (1) and stable Eurasia; (2) and (3) K1 and K2, respectively (Enkin et al. [20]).

6. Conclusions

Paleomagnetic results of the Cretaceous red sandstones collected from the eastern part of the SCB lead to the following conclusions:

(1) Early and Late Cretaceous paleomagnetic poles for the northern part (Zhejiang Province) of the eastern SCB in the present study are located at almost the same positions. These paleomagnetic pole positions are indistinguishable from the Early Cretaceous paleomagnetic pole reported previously for Zhejiang [37] at a 95% confidence level.

(2) Four Cretaceous paleomagnetic poles for the three independent parts of eastern SCB from the present study are located at almost the same positions. These paleomagnetic pole positions, however, are located at separate positions from the Late Cretaceous paleomagnetic pole compiled by Gilder et al. [12] for the coastal regions including Fujian (the central part of the studied range) and Guangdong (the southern part). This difference is due to a less reliable paleomagnetic pole calculating conducted using an insufficient amount of data [12].

(3) Four Cretaceous paleomagnetic pole positions for the three independent parts of eastern SCB agree with that for Sichuan in western SCB [12] at a 95% confidence level. This observation indicates that there is no relative tectonic movement between the eastern and western sides of the SCB and implies that a large part of the SCB (excluding the 400-km-wide swath along the RRF) has behaved as a coherent (stable) body since the Cretaceous.

(4) The global mean paleomagnetic pole for the coherent (stable) body of the SCB is worth using as the reference Cretaceous paleomagnetic pole of the SCB.

(5) Establishment of the reference Cretaceous paleomagnetic pole of the SCB enables the estimate of tectonic deformation within the SCB, especially in the vicinity of the RRF, more precisely, and a

discussion of the relative movements between SCB and other tectonic units. The reference Cretaceous paleomagnetic pole of SCB agrees strongly with the reference Eurasian paleomagnetic pole at the Cretaceous time, indicating no relative tectonic movement between the stable body of the SCB and stable Eurasia since the Cretaceous.

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